Cross-Generational Accommodation of Cantonese Sound Change Features Toward Human and AI Interlocutors

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Introduction

➢ **From improving AI performance to studying AI's impact on human performance:**

- **Previous research on human-AI interaction (HAI):** mostly focused on improving AI speech recognition and production capabilities
- **Little attention**: AI's influence on human speech and language development

➢ **AI: a distinct social interlocutor**

- **Human interlocutor:** existing social groups and speech communities
- **AI interlocutor:** new type of social identity-> different influence on human speech?
- **Age differences in susceptibility to AI influence** [1,2]

	Young adults	Children
Experience with AI	More extensive	Less extensive
Social perception of AI	Less likely to anthropomorphize AI	More likely to anthropomorphize AI
Expectation of AI	More realistic / calibrated	Potentially higher
Trust in AI	More selective / cautious	Potentially more trusting

Can AI become a new source of phonetic influence, and does its influence differ across generations?

➢ **The current study:**

- **Phonetic accommodation:** a window into HAI influence
  - **Variant adoption**: Examining the adoption of socially meaningful sound change features in HAI vs. **Human-Human Interaction (HHI)**
- **Research Questions:**
  - Do speakers differ in their adoption of sound change variants following interactions with AI versus human interlocutors?
  - Do these patterns of variant adoption differ between children and young adults?

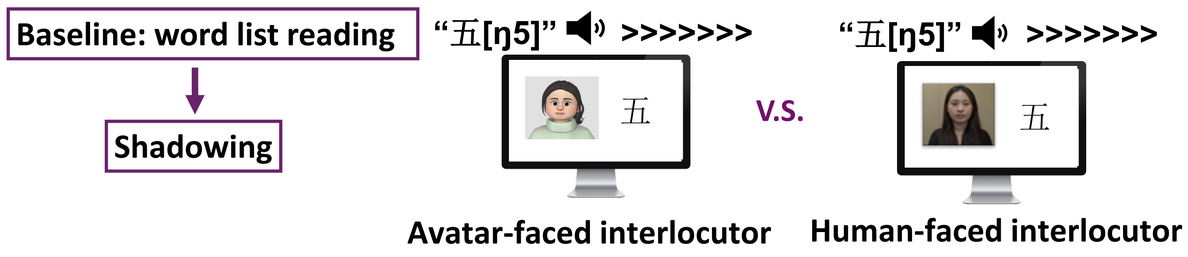
Method

Data Collection

➢ **Participant:**

- 16 native HKC-speaking young adults: mean age = 22.4±3.03 yrs
- 42 native HKC-speaking children: mean age = 10.40±0.96 yrs

➢ **Procedure:** shadowing paradigm [3]



➢ **Stimuli:**

- Monosyllabic/disyllabic syllables created based on nine HKC sound change features/phonetic variation:

(1) initial [ŋ] deletion, **(2)** initial [n]-[l] merger, **(3)** hypercorrection of initial [l], **(4)** initial [k]-[kʷ] merger, **(5)** syllabic [ŋ]-[m] merger, **(6)** [n]-[ŋ] coda merger, **(7)** palatalization of sibilants, **(8)** pinjam, **(9)** high falling of T1

- **Less frequent variants ->** target stimuli (2/3) (e.g., **[n]** in initial [n]-[l] merger: 你 [**n**ei5] 'you')
- **More frequent variants ->** filler (1/3) (e.g., **[l]** in initial [n]-[l] merger: 你 [**l**ei5] 'you')

Note: As data collection progressed, the poster incorporated data from a larger sample of children and a greater number of stimuli than those reported in the published proceeding.

Data Processing

➢ Two phonetically trained native HKC raters categorized productions as **target (1)** or **filler (0)** variants (e.g.,五[ŋ̩5]-> **target**; 五[m̩5]-> **filler**).

➢ Model: GLMM with a binomial response using lme4

- Fixed effects: Condition (baseline vs. shadowing AI face vs. shadowing human face), Age group (children vs. adults), and their interaction
- 'Emmeans' for post-hoc analysis

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Results

➢ **HHI vs. HAI**:

- **Young adults:**
  - **Significantly more likely to use target variants when shadowing** (baseline vs. HAI: est. = -2.10, p <0.001; baseline vs. HHI: est. = -2.08, p <0.001)
  - No significant evidence of differences in HHI vs. HAI
- **Children:**
  - **Significantly more likely to use target variants when shadowing** (baseline vs. HAI: est. = -1.31, p <0.001; baseline vs. HHI: est. = -1.19, p <0.0001)
  - Significantly more likely to adopt the target variants produced by the AI interlocutor vs. the human interlocutor (est. = 0.11, p = 0.04).

➢ **Children vs. Young adults**: **children** were overall less likely to adopt the interlocutors' target variants during shadowing compared to young adults (HHI: est. = -0.87, p < 0.001; HAI: est. = -0.79, p < 0.001).

➢ **Individual variation** (Figure 1. Right panel)

- Substantial individual differences in the magnitude of phonetic accommodation.
- Individuals also varied in their relative accommodation towards human and AI interlocutors. There are speakers who showed considerable large difference between shadowing AI interlocutor vs. shadowing human interlocutor (red arrows).

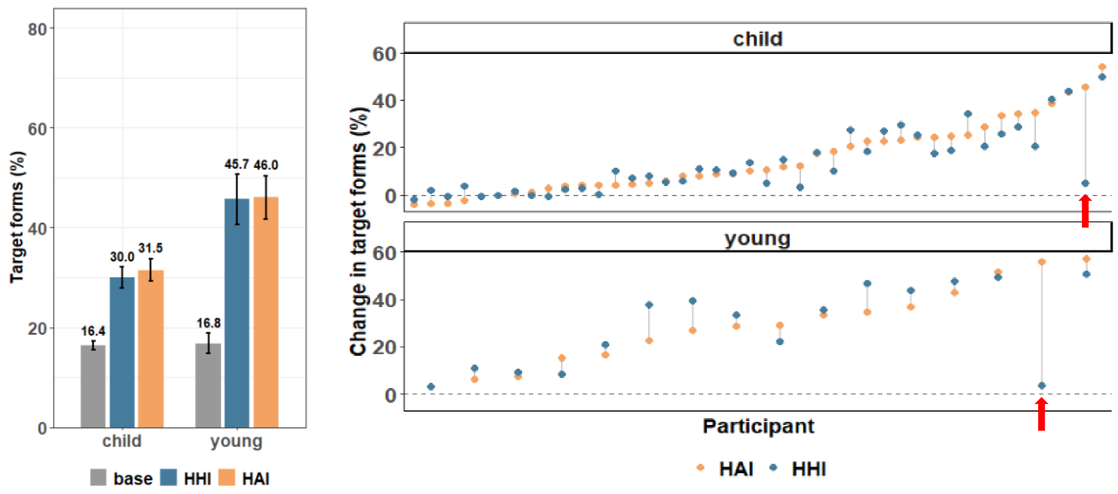


Figure 1. *(Left) Mean percentage of adopting the target forms. (Right) Individual variation in percentage change of target-form production relative to baseline. Negative value = divergence, positive value = convergence.*

Conclusion

➢ **HHI vs. HAI:**

- Both age groups accommodate to AI and human interlocutors, suggesting that **AI agents can serve as sources of phonetic influence**.
- **Children** may be more attracted to AI interlocutors than adults are.
  - AI interlocutor may be perceived as more novel and favorable stimuli. They like to interact with AI interlocutor more than young adults do.
  - AI interlocutors may represent more engaging social stimuli for children, potentially increasing their attention to the model talker's speech and facilitating phonetic uptake.

➢ **Children vs. Young adults: weaker accommodation**

- Previous studies: accommodation = gradual changes in fine-grained phonetic details [4]
- The current study:
  - Accommodation = adopting a categorically different variant -> more likely to be affected by **metalinguistic awareness** and **social skills** -> more developed for adults -> more accommodation

➢ **Individual variation**

- Phonetic accommodation in HAI is not a uniform response but may depend on individual susceptibility and sensitivity to interlocutor identity.

➢ **Ongoing work:**

- Investigating young adults' phonetic accommodation when interacting with human-voice versus AI-voice interlocutors.
- Development of AI-directed speech registers across interactions with AI interlocutors presented through voice and video.

References: [1] Flanagan T, Georgiou NC, Scassellati B, Kushnir T. School-age children are more skeptical of inaccurate robots than adults. Cognition. 2024 Aug;249:105814. **[2]** Hoehl S, Krenn B and Vincze M. Honest machines? A cross-disciplinary perspective on trustworthy technology for children. Front. Dev. Psychol. 2024. 2:1308881. **[3]** Goldinger SD. Echoes of echoes? An episodic theory of lexical access. Psychol Rev. 1998 Apr;105(2):251-79. [**4]** K. Nielsen, "Phonetic imitation by young children and its developmental changes," Journal of Speech, Language, and Hearing Research, vol. 57, no. 6, pp. 2065–2075, 2014.